Airline Flight Price & Performance Analytics | Power BI

🎯 Business Problem

Airline ticket pricing varies significantly depending on the **airline, route, class, number of stops, booking lead time, travel duration and departure/arrival schedule**.

For travelers and airline businesses, simply looking at individual ticket prices doesn't provide enough information to understand overall pricing patterns.

This project was designed to answer questions such as:

- Which airlines have the highest ticket prices?
- Which airline has the lowest-priced flights?
- Which cities have shorter flight durations?
- Which class is more common?
- How many flights have direct vs. connecting stops?
- Which arrival/departure times have the highest flight volume?
- How does booking lead time affect the available flights?
- Which airlines dominate different cities?
- What are the cheapest and most expensive flight options?

The objective was to transform **300K+ flight records** into an interactive Power BI dashboard for easier airline and pricing analysis.

📂 2. Dataset Overview

The Excel dataset contains:

300,153 flight records

12 analytical fields

Field	Description
Index	Flight record identifier
Airline	Airline company
Flight	Flight number
Source_City	Departure city
Departure_Time	Departure time category
Stops	Number of stops
Arrival_Time	Arrival time category
Destination	Destination city
Class	Economy / Business

Field	Description
Duration	Flight duration
Days_Left	Days remaining before flight
Price	Ticket price

This structure makes the dataset suitable for **airline pricing, route and booking-behavior analysis**.

🔄 3. Project Workflow

Step 1 — Data Exploration

I first examined the flight dataset to understand:

- Airline distribution
- Route patterns
- Ticket prices
- Flight duration
- Number of stops
- Travel class
- Departure and arrival periods
- Days left before travel

The dataset contains **300,153 flight records**, making it large enough to identify meaningful pricing and operational patterns.

Step 2 — Data Preparation

The data was organized into analytical dimensions:

Airline → Route → Class → Stops → Duration → Days Left → Price

This allowed the dashboard to analyze ticket pricing from multiple perspectives.

Step 3 — KPI Development

The dashboard highlights five headline metrics:

KPI	Dashboard Value
Total Flight Records	**300.15K**
Last Flight	**UK-996**
Cheapest Price	**₹1,105**
Last Destination	**Mumbai**

KPI	Dashboard Value
Cheapest Class	**Economy**

The **₹1,105 minimum ticket price** is also confirmed directly from the underlying dataset.

📊 4. Key Dataset Insights

✈ 1. Vistara dominates the dataset

Airline record distribution:

Airline	Records
Vistara	**127,859**
Air India	80,892
Indigo	43,120
GO_FIRST	23,173
AirAsia	16,098
SpiceJet	9,011

Vistara represents the largest share of flight records in the dataset, followed by Air India.

This means Vistara has the strongest representation in the available flight inventory.

💰 5. Vistara has the highest maximum ticket price

The maximum ticket prices by airline show:

Airline	Maximum Price
Vistara	**₹123,071**
Air India	₹90,970
SpiceJet	₹34,158
GO_FIRST	₹32,803
Indigo	₹31,952
AirAsia	₹31,917

Vistara therefore has both a very large number of records and the **highest maximum ticket price** in the dataset.

The dashboard visual titled **"MAX Price Over Airline"** highlights this pattern.

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6. Cheapest ticket

The lowest ticket price in the dataset is:

₹1,105

This flight is:

- **Airline:** Indigo
- **Flight:** 6E-6113
- **Source:** Chennai
- **Destination:** Hyderabad
- **Class:** Economy
- **Stops:** One
- **Duration:** 5.92 hours
- **Days Left:** 31

This provides a useful example of how a low-cost ticket can occur even on a connecting flight.

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7. Economy dominates the dataset

Class distribution:

Class	Records	Approx. Share
Economy	**206,666**	**68.9%**
Business	93,487	31.1%

The dashboard therefore shows approximately:

Economy → 69%

Business → 31%

Economy is clearly the dominant class in the dataset.

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8. One-stop flights dominate

Stops distribution:

Stops	Records	Approx. Share
One	**250,863**	**83.6%**
Zero	36,004	12.0%
Two or More	13,286	4.4%

This matches the dashboard's **No. of Stops Over Flight** visual:

- **One stop → 84%**
- **Zero stops → 12%**
- **Two or more → 4%**

So the overwhelming majority of flights in this dataset have exactly **one stop**.

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9. Night is the most common arrival period

Arrival distribution:

Arrival Time	Flights
Night	**91,538**
Evening	78,323
Morning	62,735
Afternoon	38,139
Early Morning	15,417
Late Night	14,001

Therefore, **Night** is the most common arrival period.

The dashboard's **No Of Flight Over Arrival** chart reflects this pattern.

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10. City-wise flight volume

Source-city distribution:

City	Flights
Delhi	**61,343**
Mumbai	60,896
Bangalore	52,061
Kolkata	46,347
Hyderabad	40,806
Chennai	38,700

Delhi has the largest number of source-city flight records, narrowly ahead of Mumbai.

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11. Minimum duration varies by city

Minimum flight duration found in the dataset:

City	Minimum Duration
Chennai	**0.83 hrs**
Bangalore	**0.83 hrs**
Hyderabad	0.92 hrs
Mumbai	1.17 hrs
Delhi	1.92 hrs
Kolkata	2.08 hrs

This is consistent with the dashboard's **Min Duration Over City** visualization.

📅 12. Booking lead time

The dashboard includes **Days Left** as an interactive filter.

This is an important variable because airline ticket prices can vary according to how far in advance a flight is booked.

A useful future enhancement would be:

> **Average Price vs. Days Left**

This would allow the dashboard to directly show whether prices increase as the travel date approaches.

🧠 13. Overall Business Story

The project reveals a clear picture of the airline market represented by the dataset:

> **The dataset contains more than 300K flight records, with Vistara having the largest representation. Economy flights account for approximately 69% of records, while one-stop flights dominate at approximately 84%. Night is the most common arrival period, and Delhi and Mumbai are the two largest source cities. Vistara also has the highest maximum ticket price at ₹123,071, while the cheapest available ticket in the dataset is ₹1,105.**

This makes the project useful for both **traveler decision-making and airline pricing analysis**.

💡 14. Business Recommendations

✈ 1. Analyze price by booking lead time

Create:

Days Left vs Average Price

to understand how ticket prices change as departure approaches.

💰 2. Compare airline pricing

Analyze:

Airline → Average Price → Median Price → Maximum Price

instead of relying only on maximum price.

🏷️ 3. Compare Economy vs Business

Calculate:

Average Economy Price vs Average Business Price

for every airline and route.

🛑 4. Evaluate stops vs pricing

Analyze whether:

Non-stop → One-stop → Two+ stops

results in significantly different ticket prices.

🗺️ 5. Route-level analysis

A future dashboard could include:

Source City → Destination → Average Price → Duration

to identify expensive and affordable routes.